Assignment 3: Transformer is All You Need

Tiny Shakespeare Next-Token Model (From-Scratch Transformer)

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Code: https://github.com/zz3231/COMS4995-Applied-Machine-Learning-HW/tree/main/HW3

1 Introduction

This report presents a **minimal causal Transformer** built from scratch in PyTorch for **next-token prediction** on Tiny Shakespeare ($\sim$1.1 M chars). The training objective is autoregressive language modeling: maximize $\log p(x_t \mid x_{<t})$ via cross-entropy. We report PPL $= \exp(\mathrm{CE}_{\mathrm{val}})$; lower is better.

The architecture uses two Transformer blocks, $d_{\mathrm{model}} = 128$, eight heads, FFN width 512, RMSNorm, sinusoidal PE, and causal self-attention. Tokenization is BPE with vocab ≤ 500, fit on training text only. Beyond the main run we present a **PE ablation**, a **shorter-context** study, and qualitative generation.

2 Methods

2.1 Data and tokenization

The first 80% of characters form `train_text`; the rest is `val_text`. A ByteLevel BPE tokenizer is trained on `train_text` alone (`vocab_size`=500, `min_frequency`=2). Overlapping windows of **65** tokens with stride **32** yield $\sim$12.8k training sequences, each providing 64 supervised next-token positions.

2.2 Architecture

Pre-LN blocks: $x \leftarrow x + \mathrm{Drop}(\mathrm{Sublayer}(\mathrm{RMSNorm}(x)))$ for both attention and FFN. Causal MHA computes $\mathrm{softmax}(QK^\top/\sqrt{d_k} + M)\,V$ where M is $-\infty$ above the diagonal. Sinusoidal PE is added to embeddings scaled by $\sqrt{d_{\mathrm{model}}}$. A final RMSNorm + linear head produces vocabulary logits ($\sim 10^6$ params total).

2.3 Training

AdamW (lr 3×10^{-4}, wd 0.01), batch 32, dropout 0.1, gradient clip 1.0, 25 epochs, fixed seed. Validation uses `torch.no_grad()`.

2.4 Experiments

(A) PE ablation: same architecture without sinusoidal PE, 15 epochs. Figure 4 aligns the first 15 epochs of the main model against the full no-PE run for epoch-matched comparison. **(B) Shorter context:** block 33 / stride 16 (32 positions), 12 epochs. Not directly comparable to the main model (different T, windows, epochs, initialization); probes the $O(T^2)$ compute tradeoff.

3 Results

3.1 Training curves and perplexity

Figure 1 shows CE and PPL over 25 epochs. Train CE drops steeply (5.06 $\rightarrow$ 3.74 by epoch 5) then decelerates to **3.38**; val CE plateaus earlier at **3.72** (PPL = **41.33**). The widening train–val gap after epoch $\sim$5 signals mild overfitting: the model memorizes training n-grams faster than it generalizes, though the gap stays modest (0.34 nats by epoch 25). PPL improves most dramatically in the first five epochs (83.5 $\rightarrow$ 48.6) with diminishing returns thereafter.

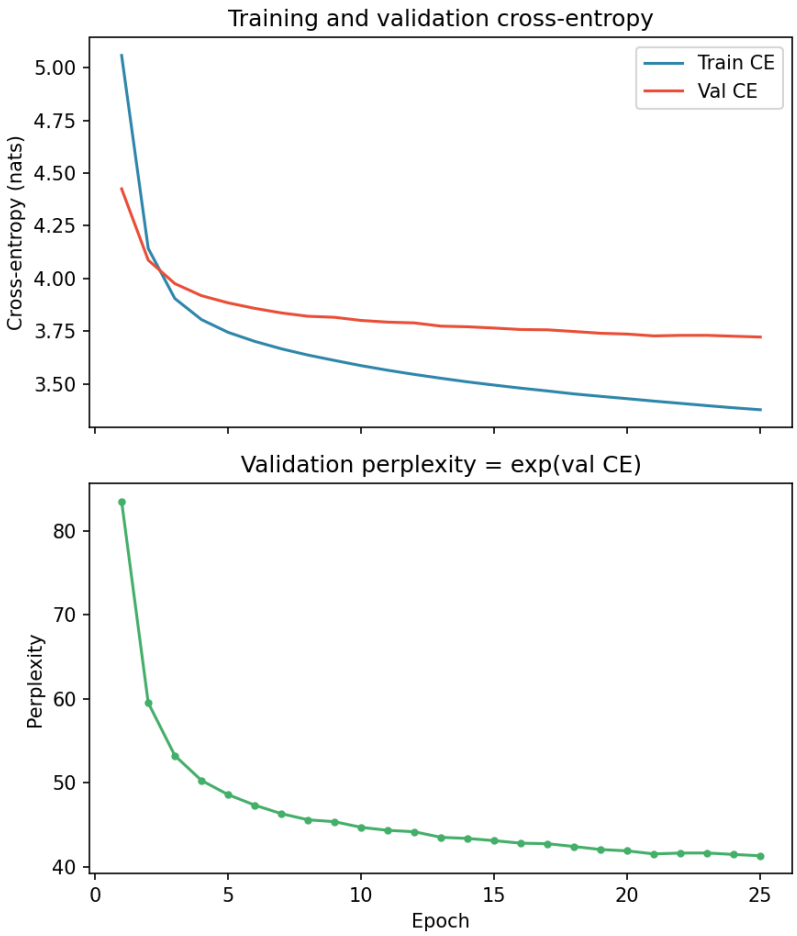

Figure 1: Train/val CE (top) and val PPL (bottom). The train–val gap widens after epoch 5.

Table 1: Hyperparameters and final metrics.

Setting	Value
Vocabulary	500 BPE (train-only)
Block / stride (main)	65 / 32 (64 positions)
d_{model}, layers, heads, d_{ff}	128, 2, 8, 512
Optimizer	AdamW, lr= 3×10^{-4}, wd= 0.01, clip= 1.0
Main: 25 ep. / val CE / PPL	3.72 / **41.33**
No-PE: 15 ep. / val PPL	**43.58**
Short ctx: blk 33, 12 ep. / PPL	**40.19**

3.2 Attention visualization and analysis

Figures 2–3 show causal attention for a validation prefix that reads approximately *"you are, That is, a woman; if you be more, you're none; If you be"* (each axis cell is one BPE token). Both maps are from the **converged** model (epoch 25); the strictly **lower-triangular** structure confirms the causal mask.

Layer 0, head 0 (Figure 2).

1. **Diagonal / recency bias.** Early tokens ("y", "ou", "are") place most weight on themselves or their immediate predecessor, forming a bright band along the diagonal—typical of lower-layer heads that propagate local subword identity into the residual stream.
2. **Content-anchored links.** "om" (from "woman") attends sharply to "are" (weight ≈0.9), suggesting the model has learned a distributional association between the copula and its noun complement. The

semicolon at position 12 focuses on the preceding "if", linking the conditional clause head.

3. **Diffuse spread for later queries.** Positions with $\geq$15 available keys show broad, light shading across many past tokens rather than sharp peaks—this head mixes wide context for downstream layers.

Layer 1, head 2 (Figure 3). Compared to L0H0, this deeper head shows **sparser, more peaked** attention:

1. **Long-range "you"-tracking.** "is" (pos. 6) attends strongly back to "ou" (pos. 2, part of "you")—a four-token span. "more" (pos. 16) concentrates weight $\approx$0.7 on "you" (pos. 14); "re" (pos. 20, part of "you're") reaches back to "you" as well. This head appears to **track the pronoun "you"** across the sequence, a rudimentary subject-tracking role.
2. **Punctuation as clause anchors.** Commas and semicolons receive concentrated attention from later queries, serving as positional landmarks for clause boundaries.
3. **Head diversity.** Comparing the same prefix across two heads reveals clear functional differentiation: L0H0 favors local, diffuse mixing; L1H2 favors sparse, long-range content-specific links. This division of labor mirrors findings in larger Transformer LMs and shows that multi-head attention learns complementary features even at small scale.

Limitation. Heatmaps are from the converged model only; we did not render attention at early epochs. However, the steep initial PPL drop suggests attention specialization likely emerges within the first few epochs.

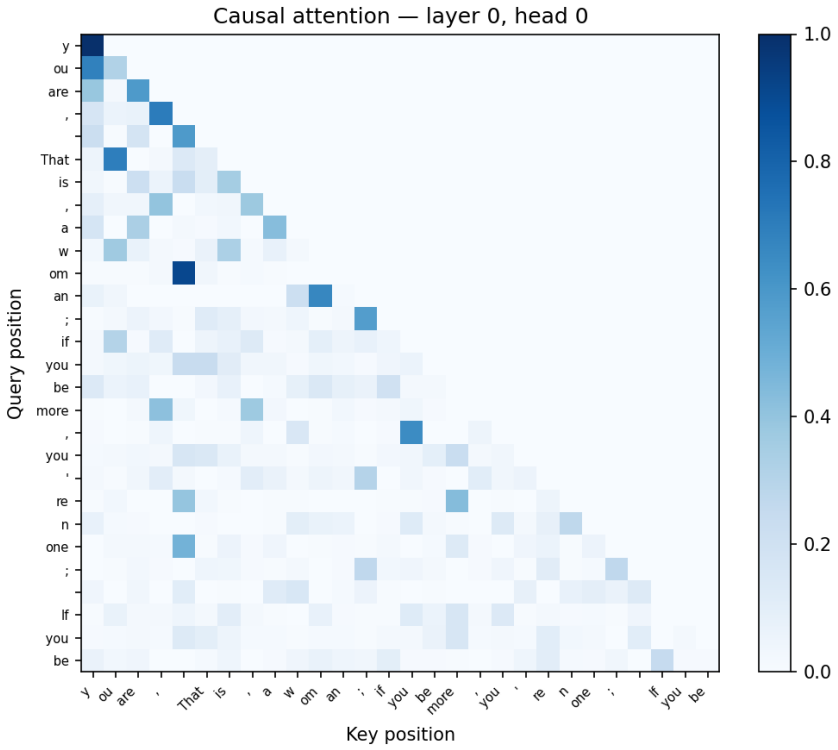

Figure 2: Layer 0, head 0: note the diagonal recency band and the "om" $\rightarrow$ "are" link.

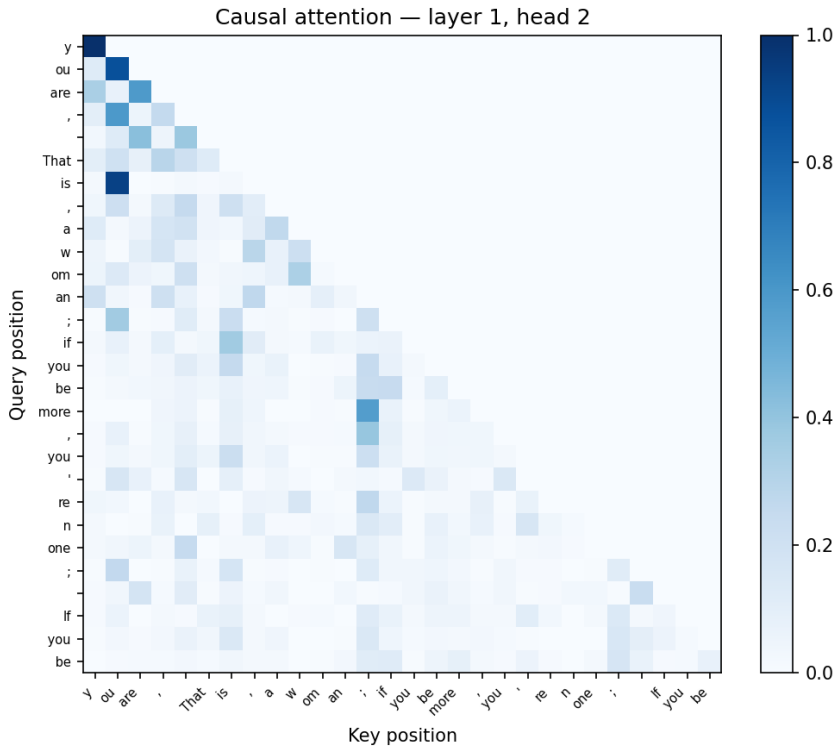

Figure 3: Layer 1, head 2: same prefix; sparser peaks, long-range "you"-tracking.

3.3 Positional encoding ablation

At epoch 15 the main model reaches PPL **43.13** while the no-PE model reaches **43.58**—a gap of 0.45 (Figure 4). The PE model is lower at **nearly every epoch**, confirming a systematic benefit. The gap is small because (i) the corpus has strong local statistical cues (capitalized line-starts, newline patterns) that partially encode position even without PE, and (ii) with only $T = 64$, positional ambiguity is limited. Over the full 25-epoch main run PPL drops further to 41.33; since the no-PE run was capped at 15 epochs, we cannot attribute the entire 2.25-point difference to PE.

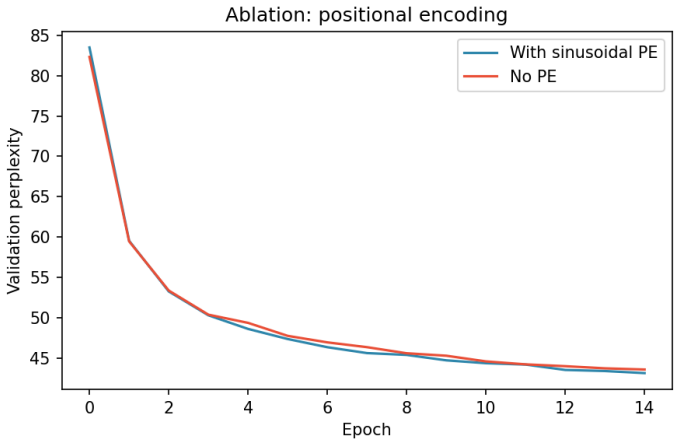

Figure 4: PE ablation: the PE model (blue) is consistently below no-PE (red) over 15 epochs.

3.4 Shorter-context experiment

With block 33 / stride 16 the model starts at PPL ∼63 and reaches **40.19** by epoch 12 (Figure 5). Starting PPL is lower than the main model's ∼83 because shorter windows average over fewer conditioning tokens,

making the per-position prediction "easier." The key takeaway is **operational**: halving T ($64 \to 32$) reduces the attention tensor from $B \times H \times 64 \times 64$ to $B \times H \times 32 \times 32$—a $4\times$ savings in attention memory and FLOPs—at the cost of losing long-range dependencies beyond 32 tokens.

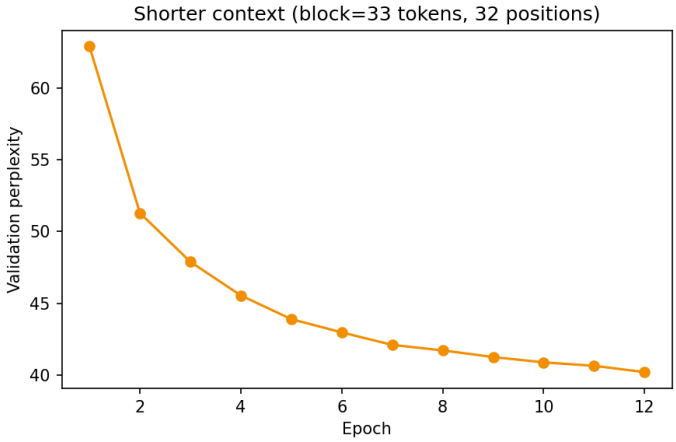

Figure 5: Shorter context (block 33): validation PPL over 12 epochs.

3.5 Qualitative generation

Temperature-0.85 sampling from the trained model produces Shakespeare-*flavored* but incoherent text: *"you are, That is, a woman; if you be more, you'er's thereock, For thought, prifft being in the dieves. . . "*. The model captures comma-separated clause structure, pseudo-archaic pronouns ("Ay", "Ty"), and stage-direction labels, but hallucinates many subwords ("thereock", "prifft"). This is expected given $\sim 10^6$ params, 500-token vocab, and 25 epochs of training.

4 Discussion

Attention patterns. The two visualized heads (§3.2) show complementary roles: L0H0 acts as a *local mixer* (diagonal recency, diffuse spread), while L1H2 is a *content pointer* (sparse "you"-tracking, punctuation anchoring). This early-layer-local / later-layer-global stratification mirrors findings in large-scale LMs and demonstrates that functional head specialization emerges even in a two-layer model.

Hyperparameter stability. Learning rate was the most sensitive knob: values above $\sim 10^{-3}$ caused loss spikes, below $\sim 10^{-4}$ slowed convergence. Gradient clipping (norm 1.0) prevented the occasional explosion that AdamW alone could not absorb. Batch size 32 and weight decay 0.01 were stable defaults.

Positional encodings. Self-attention without PE is permutation-invariant: it treats any token permutation identically. Sinusoidal PE breaks this symmetry; our ablation confirms a consistent but small benefit. RoPE (rotary PE) is an alternative that injects position into Q, K products rather than additive embeddings, potentially offering better length extrapolation.

Runtime bottlenecks. The dominant cost is the $O(BHT^2)$ attention score tensor. The short-context experiment directly demonstrates the $4\times$ savings from halving T. Modern solutions include FlashAttention (fused softmax kernel, avoids materializing $T \times T$) and KV caching at inference (avoids recomputing past keys/values during autoregressive generation); neither was implemented here.

Course context. This assignment performs **supervised** next-token prediction with a fixed CE loss. This contrasts with RLHF/PPO (L11), where a policy gradient optimizes a learned reward model trained on human preferences—our work sits on the "pretraining" side of that divide.

5 AI Tool Usage Disclosure

Chat-based AI tools helped with notebook scaffolding, plotting boilerplate, and a first-draft LaTeX template. I **implemented** the model components (RMSNorm, causal MHA, FFN), **ran all experiments** on Colab GPU, **inspected every figure** to write figure-grounded analysis (e.g., the "om"$\to$"are" and "you"-tracking

observations in §3.2), **verified** all reported numbers against notebook output, and **revised** the report text for accuracy. Figures are the notebook's saved `fig_*.png` files without modification.